

- CO5 a. Explain with proper analysis of Selective Approach System of control.
 b. The XYZ manufacturing company has determined from an analysis of its accounting and production data for part number 625, that its cost to purchase is Rs. 36/- per order and Rs. 2/- per part. Its inventory carrying charge is 18% of the average inventory cost. The demand for this part is 10,000 units per annum. What is the economic order quantity and what is the optimal number of days supply per optimum order.
 c. Define strictly deterministic, strictly non-deterministic and fair games. Find the range of values of p and q which will render entry $(2, 2)$ a saddle point for the game.

	B_1	B_2	B_3
A_1	2	q	6
A_2	p	5	10
A_3	6	3	4

2+5

- CO6 a. In a departmental store, one cashier is there to serve the customers. And the customers pick up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the cashier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate, find: i) Average number of customers in the queue or average queue length. ii) Average time a customer spends in the system. iii) Average time a customer waits before being served.

- b. What are the advantages of PERT and CPM networks?
 What is the significance of cost slope?

- c. A company manufacturing plant and equipment for chemical processing is in the process of quoting tender called by public sector undertaking. Help the manager to find the project completion time to participate in the tender.
 i) Perform forward and backward pass. ii) Identify the critical path and project completion time. iii) If activity G is delayed by 2 Days, what happens in project completion time change and/or critical path change?

Activities	Predecessor	Days
A	—	3
B	—	4
C	A	5
D	A	6
E	C	7
F	D	8
G	B	9
H	E, F, G	3

3

8

2+2+2

M.M

Time: 1hr. Set-1 CT-2, 2026 BE-IT 2nd Year 1st Semester Subject: Mathematics for IT-II (IT/BS/B/T/222) Full Marks: 30

CO5	a. Define with example of replenishment cost, procurement cost, and stock-out-cost related to inventory control.	<table><tr><td></td><td>M_1</td><td>M_2</td><td>M_3</td><td>M_4</td><td>M_5</td></tr><tr><td>J_1</td><td>4</td><td>2</td><td>1</td><td>7</td><td>-1</td></tr><tr><td>J_2</td><td>1</td><td>-4</td><td>-6</td><td>-7</td><td>6</td></tr><tr><td>J_3</td><td>3</td><td>2</td><td>3</td><td>4</td><td>2</td></tr><tr><td>J_4</td><td>-6</td><td>1</td><td>-1</td><td>0</td><td>4</td></tr><tr><td>J_5</td><td>0</td><td>0</td><td>6</td><td>0</td><td>0</td></tr></table>		M_1	M_2	M_3	M_4	M_5	J_1	4	2	1	7	-1	J_2	1	-4	-6	-7	6	J_3	3	2	3	4	2	J_4	-6	1	-1	0	4	J_5	0	0	6	0	0	3
			M_1	M_2	M_3	M_4	M_5																																
J_1	4	2	1	7	-1																																		
J_2	1	-4	-6	-7	6																																		
J_3	3	2	3	4	2																																		
J_4	-6	1	-1	0	4																																		
J_5	0	0	6	0	0																																		
	b. A shopkeeper has a uniform demand of an item at the rate of 50 units per month. He buys from supplier at the cost of Rs. 6/- per item and the cost of ordering is Rs. 10/- each time. If the stock holding costs are 20% per year of stock value, how frequently should he replenish his stocks? What is the optimal cost of inventory and what is the total cost?		3																																				
	c. Define payoff matrix and saddle point. Apply max-min and min-max principle to solve the game (pay-off matrix $J_i \rightarrow M_j$):		2+5																																				
CO6	a. At a public telephone booth in a post office arrivals are considered to be Poisson with an average inter-arrival time of 12 minutes. The length of phone call may be assumed to be distributed exponentially with an average of 4 minutes.	<table><tr><th>Activities</th><th>Duration (days)</th><th>Predecessor</th></tr><tr><td>A</td><td>3</td><td>—</td></tr><tr><td>B</td><td>5</td><td>A</td></tr><tr><td>C</td><td>4</td><td>A</td></tr><tr><td>D</td><td>6</td><td>B</td></tr><tr><td>E</td><td>3</td><td>B</td></tr><tr><td>F</td><td>5</td><td>C</td></tr><tr><td>G</td><td>4</td><td>D, E</td></tr><tr><td>H</td><td>2</td><td>F</td></tr><tr><td>I</td><td>3</td><td>G, H</td></tr></table>	Activities	Duration (days)	Predecessor	A	3	—	B	5	A	C	4	A	D	6	B	E	3	B	F	5	C	G	4	D, E	H	2	F	I	3	G, H	2+2+2						
	Activities		Duration (days)	Predecessor																																			
A	3	—																																					
B	5	A																																					
C	4	A																																					
D	6	B																																					
E	3	B																																					
F	5	C																																					
G	4	D, E																																					
H	2	F																																					
I	3	G, H																																					
	i) What is the probability that a fresh arrival will not have to wait for phone?		3 8																																				
	ii) What is the average length of queues that form time to time?																																						
	iii) What is the fraction of time that the phone is busy?																																						
	b. Define with example of optimistic time, pessimistic time, and slack time.																																						
	c. A project consists of the following activities is given below.																																						
	i) Find the earliest, latest and slack time of each activity.																																						
	ii) Hence find the critical path and project duration.																																						

- CO5
- Give a graphical interpretation of EOQ. Measure the EOQ without shortage and deterministic uniform demand.
 - A stockiest has to supply 400 units of a product every Monday to his customers. He gets the product at Rs. 50/- per unit from the manufacturer. The cost of ordering and transportation from the manufacturer is Rs. 75 per order. The cost of carrying inventory is 7.5% per year of the cost of the product. Determine the economic lot size and the total optimal cost (including the capital cost).
 - How will you solve a game without any saddle point? For what value of λ , the game is strictly determinable.

	B_1	B_2	B_3
A_1	λ	7	3
A_2	-2	λ	-8
A_3	-3	4	λ

3

3

2+5

- CO6
- Customers arrive at one window drive in a bank according to a Poisson distribution with a mean of 10 per hour. Service time per customer is exponential with a mean of 5 minutes. The space in front of the window, including that for the serviced car can accommodate a maximum of three cars. Other cars can wait outside the space. i) How long an arriving customer is expected to wait before starting service? ii) What is the probability that an arriving customer can drive directly to the space in front of the window and will have to wait outside the indicated space?
 - How the CPM type of networks differ from the PERT networks? What is crash time?
 - A small project is composed of 7 activities whose time estimates are listed below. Activities being identified by their beginning (i) and ending (j) node numbers. i) Find the expected project completed time. ii) Calculate the probability that the project will be completed at least 3 weeks than expected. iii) If the project is completed in 15 weeks, what is the probability of not meeting the due date?

Activities		Time in weeks			
i	j	t_o	t_p	t_s	
1	2	1	1	7	
1	3	1	4	7	
1	4	2	2	8	
2	5	1	1	1	
3	5	2	5	14	
4	6	2	5	8	
5	6	3	6	15	



3

8

14